Manipulating string and character data

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The string concatenation operator

```
SELECT
  first_name,
  last_name,
  first_name || ' ' || last_name AS full_name
FROM customer
```

```
+------------+-----------+------------------+
| first_name | last_name | full_name        |
|------------|-----------|------------------|
| MARY       | SMITH     | MARY SMITH       |
| LINDA      | WILLIAMS  | LINDA WILLIAMS   |
+------------+-----------+------------------+
```

Or with CONCAT() function

String concatenation with a non-string input

```
SELECT
  customer_id || ': '
  || first_name || ' '
  || last_name AS full_name
FROM customer;
```

```
+------------------+
| full_name        |
|------------------|
| 1: MARY SMITH    |
| 2: LINDA WILLIAMS |
+------------------+
```

Changing the case of a string
UPPER(), LOWER() - uppercase, lowercase
INITCAP() - titlecase

Replacing characters in a string

```
SELECT
    REPLACE(description, 'A Astounding',
              'An Astounding') as description
FROM film;
```

```
+-------------------------------------------------------+
| description                                           |
|-------------------------------------------------------|
| A Epic Drama of a Feminist And a Mad Scientist...     |
| An Astounding Epistle of a Database Administrator...  |
| An Astounding Reflection of a Lumberjack And a Car...  |
+-------------------------------------------------------+
```

Manipulating string data with REVERSE

```
SELECT
  title,
  REVERSE(title)
FROM
  film AS f;
```

```
+-------------------------------------+
| title            | reverse(title)   |
|-------------------------------------|
| ACADEMY DINOSAUR | RUASONID YMEDACA |
| ACE GOLDFINGER   | REGNIFDLOG ECA   |
+-------------------------------------+
```

CHAR_LENGTH() or LENGTH() returns an integer of the length of a string

Finding the position of a character in a string

```
SELECT
    email,
    POSITION('@' IN email)
FROM customer;
```

```
+-------------------------------------+------------------------+
| email                               | POSITION('@' IN email) |
|-------------------------------------|------------------------|
| MARY.SMITH@sakilacustomer.org       | 11                     |
| PATRICIA.JOHNSON@sakilacustomer.org | 17                     |
| LINDA.WILLIAMS@sakilacustomer.org   | 15                     |
+-------------------------------------+------------------------+
```

We can use STRPOS(email, '@') instead, it is analogous

Parsing string data

```
SELECT
    RIGHT(description, 50)
FROM film;
```

```
+----------------------------------------------------+
| description                                        |
|----------------------------------------------------|
|  who must Battle a Teacher in The Canadian Rockies |
| nd a Explorer who must Find a Car in Ancient China |
| Car who must Sink a Lumberjack in A Baloon Factory |
+----------------------------------------------------+
```

LEFT() does the same but from the left

Extracting substrings of character data

```
SELECT
    SUBSTRING(description, 10, 50)
FROM
  film AS f;
```

```
+----------------------------------------------------+
| description                                        |
|----------------------------------------------------|
| ama of a Feminist And a Mad Scientist who must Bat |
| ing Epistle of a Database Administrator And a Expl |
| ing Reflection of a Lumberjack And a Car who must  |
+----------------------------------------------------+
```

SUBSTRING(source, starting point, length)

Extracting substrings of character data

```
SELECT
    SUBSTRING(email FROM 0 FOR POSITION('@' IN email))
FROM
  customer;
```

```
+----------------------------------------------------+
| SUBSTRING(email FROM 0 FOR POSITION('@' IN email)) |
|----------------------------------------------------|
| MARY.SMITH                                         |
| PATRICIA.JOHNSON                                   |
| LINDA.WILLIAMS                                     |
+----------------------------------------------------+
```

Extracting substrings of character data

```
SELECT
    SUBSTRING(email FROM POSITION('@' IN email)+1 FOR CHAR_LENGTH(email))
FROM
  customer;
```

```
+----------------------------------------------------------------------+
| SUBSTRING(email FROM POSITION('@' IN email)+1 FOR CHAR_LENGTH(email)) |
|----------------------------------------------------------------------|
| sakilacustomer.org                                                   |
| sakilacustomer.org                                                   |
| sakilacustomer.org                                                   |
+----------------------------------------------------------------------+
```

SUBSTR() is analogous, but only allows the comma-separated arguments, not the FROM...

Removing whitespace from strings

```
TRIM([leading | trailing | both] [characters] from string)
```

- **First parameter:** [leading | trailing | both]
- **Second parameter:** [characters]

Optional parameters, default is both and a blank space respectively
There is LTRIM() and RTRIM() too, from left and right, leaves the opposite

Padding strings with character data

```
SELECT LPAD('padded', 10, '#');
```

```
+-------------+
| LPAD        |
```

```
|-------------|
| ####padded  |
+-------------+
```

RPAD() is analogous just from the right
By default it is padded with a space character
If the length is less than the original length of the string, it is going to be truncated

```sql
SELECT
  UPPER(c.name) || ': ' || f.title AS film_category,
  -- Truncate the description without cutting off a word
  LEFT(description, 50 -
    -- Subtract the position of the first whitespace character
    POSITION(
      ' ' IN REVERSE(LEFT(description, 50))
    )
  )
FROM
  film AS f
  INNER JOIN film_category AS fc
    ON f.film_id = fc.film_id
  INNER JOIN category AS c
    ON fc.category_id = c.category_id;
```

Run Code | Submit Answer

query result | film | category | film_category

film_category	left
ACTION: WEREWOLF LOLA	A Fanciful Story of a Man And a Sumo Wrestler who
ACTION: WATERFRONT DELIVERANCE	A Unbelieveable Documentary of a Dentist And a
ACTION: UPRISING UPTOWN	A Fanciful Reflection of a Boy And a Butler who